

Learning-Augmented Aggregate Bounds over Hidden Matchings and Privacy-Coarsened Attributes

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ABSTRACT

Public releases often preserve one row per transaction while suppressing the key that joins rows into events and coarsening sensitive endpoint attributes. Downstream aggregate queries then depend jointly on a latent relation and the attributes attached to its endpoints. We represent the release by feasible worlds that jointly assign a nonbipartite matching and categorical endpoint attributes subject to core–buffer boundary rules, released supports, and global count constraints. We report attained query endpoints rather than one imputed graph. Independent attribute supports collapse to ordinary weighted matching, whereas feasibility under an explicit paired threshold-release operator is NP-complete even on a complete bipartite outer graph. Learning enters through an additive matching score whose positive-affine-invariant regret radius is calibrated on independent markets with observed relations. Under market-level exchangeability and candidate support, the restricted feasible-world set, and hence every downstream query range computed from it, contains the true world with finite-sample probability at least $1 - \alpha$. In a fixed six-pair synthetic benchmark, an illustrative tight radius for a deliberately query-leaking scorer reduces range width by 95.8% but retains only 18/120 true matchings. Calibration retains 117/120 and covers the downstream statistic in 120/120, while width reduction falls to 7.6%; this is a calibration stress test, not evidence about another domain. We develop Chicago ride pooling as a flagship application in which realized co-presence is reported but partner identity and fine geography are suppressed. Complete-day Chicago estimates remain gated on verification of the current release operator and are not claimed in this working draft.

CCS CONCEPTS

• Computing methodologies → Machine learning; • Information systems → Data provenance; • Theory of computation → Design and analysis of algorithms.

KEYWORDS

uncertain linkage, hidden matching, conformal prediction, privacy coarsening

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1 INTRODUCTION

Many public datasets preserve outcomes for individual records but remove the key identifying which records participated in the same event. Sensitive endpoint attributes may also be generalized or suppressed. Aggregate quantities such as the composition of matched

pairs, referral chains, or shared-resource groups then depend on two latent objects at once: the relation and the attributes attached to its endpoints. Selecting one linkage or deleting rows with missing context creates precision by replacing released uncertainty with analyst certainty.

Propagating linkage uncertainty alone does not resolve this problem. Candidate edges can interact with latent endpoint attributes, privacy bucket counts make attribute completions mutually dependent, and the hidden matching determines which endpoint attributes enter the query. Separate link and attribute bounds can therefore combine marginally admissible choices that no single released database could have produced.

We represent the release by a set of feasible worlds. Each world jointly specifies a hidden matching and latent endpoint attributes, satisfies every released value and global coarsening constraint, and covers every core record exactly once. We compute attained lower and upper endpoints of a declared aggregate query over this set. These endpoints are conditional on the candidate supergraph and observation operator; they are not a point reconstruction, and the closed interval between them need not consist entirely of attainable values.

Learning is optional and coverage-audited. An edge model supplies a compatibility score, but node outcomes alone do not identify partner probabilities. We restrict feasible worlds by matching-level normalized regret and calibrate the radius on independent markets with observed relations. A separate supergraph budget permits selected edges outside a baseline graph. Thus learned sharpening and candidate-support sensitivity remain visible rather than being folded into one asserted graph.

Chicago ride pooling motivates the full structure. Consider a boundary-complete, internally consistent cohort with a positive shared-match flag and pooled chain size two. The public record reports realized co-presence but not the partner key. Fine geography can also be suppressed. The eventual application bounds neighborhood-context composition among those customer transactions. It does not infer rider socioeconomic status, preferences, durable ties, or all shared runs.

This working version makes three completed contributions. First, it defines joint feasible-world query semantics, identifies the collapse to weighted matching, and establishes an NP-complete coupled boundary under a precisely stated release operator. Second, it gives a matching-level split-conformal restriction whose finite-sample inclusion event propagates to all predeclared downstream queries. Third, it implements an exact small-instance oracle and a numerical joint matching–label program, supplies executable counterexamples, and freezes a proxy-shift benchmark. A production-scale certified algorithm, a second relation-truth benchmark, and a complete Chicago run remain explicit KDD submission gates.

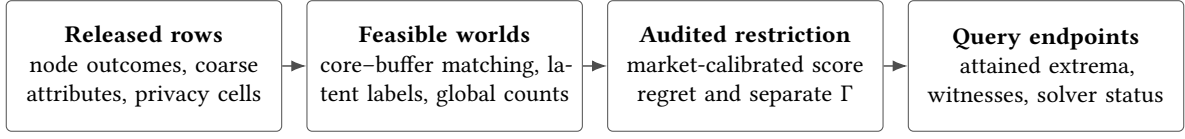


Figure 1: The method never fills in one preferred relation. The observation operator defines a joint world set; learning may restrict that set only through a separately calibrated rule; optimization then propagates all surviving ambiguity to the downstream query.

2 RELATED WORK

Uncertain linkage and aggregate queries. Globally compatible record linkages and downstream uncertainty are established objects. Sadinle models bipartite linkage as a constrained matching [8]; Hua and Pei evaluate aggregate queries over compatible probabilistic linkages [4]; and Turckapar and Krishnan compute extremal aggregate answers over candidate-constrained two-table linkages [13]. Attribute-annotated uncertain databases propagate value bounds through relational queries [3]. Our target difference is conditional and narrower: a within-file nonbipartite relation and endpoint attributes are simultaneously hidden, while global release counts can couple their completions. When attribute supports are independent, our problem collapses to ordinary weighted matching; that case is a baseline, not a novelty claim.

Weak supervision and calibrated restrictions. Multiple-instance learning recovers instance-level structure from bag labels only through modeling assumptions [6]. In our application, node match outcomes do not nonparametrically identify partner-edge rankings, and multiplying node-level noisy-OR terms is not a coherent joint distribution over overlapping matchings. We therefore treat any edge model as a scorer. Split conformal calibration supplies finite-sample marginal coverage for a set-valued prediction rule under exchangeability [5]. We apply the rank argument to complete market-level feasible worlds and then propagate the resulting inclusion event to multiple downstream aggregate queries. Candidate-graph support remains a separate condition.

Ride pooling and contextual composition. Shareability networks and dynamic assignment solve a forward allocation problem with observed requests [1, 9]. Chicago work predicts sharing and matching outcomes [7, 12], relates pooled demand to neighborhood disadvantage [11], and constructs proxy groups by spatiotemporal clustering [10]. Our flagship application asks an inverse measurement question: among a precisely defined set of realized two-transaction matches, which neighborhood-context compositions remain compatible with the suppressed partner key and the verified release operator? It does not infer rider socioeconomic status, preferences, durable ties, or an echo chamber.

3 PROBLEM AND QUERY SEMANTICS

Released records and roles. Let V index released records and let $G = (V, E)$ be a declared candidate supergraph. Core records $C \subseteq V$ belong to the target population; boundary-buffer records $B \subseteq V$ may be partners of core records; and context-only records $U = V \setminus (C \cup B)$ enter privacy counts but cannot be selected as partners.

A matching $M \subseteq E$ must satisfy

$$\deg_M(i) = 1 \quad (i \in C), \quad \deg_M(i) \leq 1 \quad (i \in B), \quad (1)$$

and contains no buffer-buffer edge. This boundary rule lets a core record use a counterpart immediately outside the target window without forcing every buffer record into the target.

Latent attributes and release constraints. Each record i has a finite support $\mathcal{A}_i$ for a latent categorical attribute a_i . Supports may be singletons for released values or larger sets for coarsened values. In the implemented reference solver, $h(i)$ assigns each record to one release cell for this modeled attribute. The verified observation operator supplies rational count bounds

$$L_{cv} \leq \sum_{i:h(i)=c} \mathbb{I}\{a_i = v\} \leq U_{cv} \quad (c, v) \in \mathcal{K}. \quad (2)$$

More specific threshold or paired-suppression logic can be compiled into the same feasible-world representation only after its scope and direction have been verified for the target release. We do not infer a converse suppression rule from missingness alone.

Feasible worlds and queries. A feasible world $F = (M, a)$ satisfies Eqs. (1)–(2), $a_i \in \mathcal{A}_i$, and any declared attribute-conditioned edge rules. Let $\mathcal{F}(O, G)$ denote the finite set of such worlds. For a symmetric pair function g , define the core-transaction query

$$Q_C(F) = \frac{1}{|C|} \sum_{\{i,j\} \in M} |\{i, j\} \cap C| g(a_i, a_j). \quad (3)$$

Core-core edges contribute twice and core-buffer edges once, so the denominator remains $|C|$. Same-bin composition uses $g(a_i, a_j) = \mathbb{I}\{b(a_i) = b(a_j)\}$; signed linear or regression queries replace g by the corresponding endpoint contribution.

We report

$$\underline{Q} = \min_{F \in \mathcal{F}(O, G)} Q_C(F), \quad \bar{Q} = \max_{F \in \mathcal{F}(O, G)} Q_C(F). \quad (4)$$

PROPOSITION 1 (ENDPOINT SEMANTICS). *If $\mathcal{F}(O, G)$ is nonempty, both extrema in Eq. (4) are attained. Every feasible world's query value lies between them. If the true world $F^\dagger$ belongs to $\mathcal{F}(O, G)$, its query value is covered. The proposition does not imply that every scalar between the endpoints is attainable.*

PROOF. The domain is a finite set of matchings and finite attribute assignments, so a real-valued query attains its minimum and maximum. The bracketing and coverage claims follow from membership. Two feasible worlds can attain only two distinct values, so finiteness alone provides no interpolation claim. $\square$

Table 1: Claim boundary induced by the four input layers.

Layer	Supplied object	Failure if misspecified
Relation support	Supergraph and core/buffer roles	True partner absent
Attribute support	Per-record finite domains	True endpoint value absent
Observation operator	Count and release factors	Inconsistent world deletion
Learned restriction	Score radius and Γ budget	Model-induced deletion

3.1 Observation Operators as Auditable Constraints

The world set must encode what the release mechanism actually implies, not a convenient completion rule. We separate four inputs. *Record supports* translate released or coarsened values into finite domains. *Count factors* constrain assignments across every row contributing to a privacy cell, including context-only rows. *Release factors* state the verified direction of the mechanism: for example, a visible value may imply that a count cleared a threshold, while a hidden value need not imply the converse. Finally, *compatibility factors* specify which attribute pairs allow an edge. An omitted rule leaves worlds in the set; it does not silently choose a default completion.

This separation yields three different statements. The score-free endpoints are conditional on the first three layers. A learned endpoint adds the fourth. A sampling statement additionally requires a design over complete markets. None of these is a confidence interval merely because two numbers are reported. The artifact therefore stores the two endpoint witnesses and a solver-status record alongside every range.

4 COUPLED ENDPOINT COMPUTATION

The method must distinguish a useful baseline from the coupled problem.

LEMMA 1 (INDEPENDENT-SUPPORT COLLAPSE). *Suppose (i) E does not depend on latent attributes, (ii) the attribute domain is Cartesian, $\prod_i \mathcal{A}_i$, with no cross-record count constraint, and (iii) the query is pair additive as in Eq. (3). Define*

$$\begin{aligned} \ell_{ij} &= \min_{a_i \in \mathcal{A}_i, a_j \in \mathcal{A}_j} |\{i, j\} \cap C|g(a_i, a_j), \\ u_{ij} &= \max_{a_i \in \mathcal{A}_i, a_j \in \mathcal{A}_j} |\{i, j\} \cap C|g(a_i, a_j). \end{aligned}$$

Then the lower endpoint is a minimum-weight core-cover matching with weights $\ell_{ij}/|C|$, and the upper endpoint is the corresponding maximum-weight matching with weights $u_{ij}/|C|$.

PROOF. For any fixed matching, selected edges share no endpoint. Under a Cartesian attribute domain, each selected edge's minimizing or maximizing endpoint assignment can therefore be chosen independently and combined into one global assignment. Labels of unselected nodes can be completed arbitrarily from their nonempty supports. Optimizing the resulting edge sums over matchings gives the stated equalities. $\square$

Thus exact cover, independent missing-bin envelopes, and signed edge objectives are weighted-matching baselines. The nonseparable problem begins when release counts or attribute-conditioned feasibility prevent the edgewise choices in Lemma 1 from coexisting.

Why the joint world changes an endpoint. Consider four core records with fixed edges ab and cd . Their supports are $\{A, B\}$, $\{A, C\}$, $\{A, D\}$, and $\{A, E\}$, respectively. Let A form bin 0, B, D bin 1, and C, E bin 2. With independent supports, the same-bin query ranges from 0 to 1: choose (B, C, D, E) for the lower witness and all A for the upper witness. Add the single global release constraint $\#\{i : a_i = A\} \leq 2$. At most one edge can now have equal labels, so the attained range contracts to $[0, 1/2]$. Optimizing a matching range and an attribute range separately would retain the impossible upper value 1.

Attributes can also change the relation rather than only its query value. Let one core c have support $\{A, B\}$ and two singleton buffers $x = A$ and $y = B$. If edge cx permits only (A, A) and cy only (B, B) , a count constraint fixing $c = A$ selects x , while one fixing $c = B$ selects y . The implemented exact oracle reproduces both examples. They isolate the interaction absent from the independent-support baseline without invoking a privacy mechanism that has not been verified.

A coupled complexity boundary. Fix an integer threshold $k \geq 3$. Consider a paired row-release operator that publishes both endpoint attributes of record i exactly when both its pickup and drop-off cells contain at least k records; otherwise it publishes neither. This is an abstract operator, not an assertion about the current Chicago release.

THEOREM 1 (PAIRED-SUPPRESSION FEASIBILITY). *For every fixed $k \geq 3$, deciding whether a feasible attribute assignment and perfect matching exist under the paired operator is NP-complete, even when the outer relation is complete bipartite, compatibility depends only on the left record's pickup attribute, and only left pickup attributes have non-singleton supports.*

PROOF SKETCH. Membership in NP follows by checking assignments, cell counts, release states, compatibility, and matching degrees. For hardness, reduce perfect three-dimensional matching on equally sized sets X, Y, Z . Each $x \in X$ chooses a latent pickup label $p_x \in Z$. For every z , add $k - 1$ visible context anchors to its pickup cell; all anchors share a separate already-large drop-off cell. Visibility forces $(k - 1) + |\{x : p_x = z\}| \geq k$. Since $|X| = |Z|$, every z is chosen exactly once. Give each x a private low-count drop-off cell so its two attributes remain paired-suppressed. Match X perfectly to Y and allow edge (x, y) under label z exactly when (x, y, z) is an input triple. Feasible worlds are therefore in bijection with perfect 3D matchings. The full construction, including the $|Z| \geq 2$ padding and size bound, is in Appendix B. $\square$

The reduction relies on paired row suppression and arbitrary tabular attribute-conditioned compatibility. The construction does not apply to endpoint-wise independent suppression and does not establish hardness for a metric-only edge rule. These boundaries must be checked, not blurred, in any application.

Threshold compilation. The threshold structure also creates an exact algorithmic foothold. Suppose an operator and every other

factor consult a release-cell count only through predicates $N_c < k$ and $N_c \geq k$; in particular, no constraint or query distinguishes counts above k .

LEMMA 2 (CAPPED-COUNT SUFFICIENCY). *Replacing every partial or final count N_c by $\tilde{N}_c = \min\{N_c, k\}$ preserves every release predicate, every feasible world, and the endpoints of every query that depends on selected relations and attributes but not on the count above k .*

PROOF. The predicates distinguish exactly the states $0, \dots, k-1$ from the single state k . The update $\tilde{N}_c \leftarrow \min\{k, \tilde{N}_c+1\}$ is therefore sufficient for every assignment sequence. It changes neither the matching nor any attribute label, so the query is unchanged. $\square$

Compile partner choices, attribute assignments, release factors, and capped count recursions into a finite factor graph. Let τ be the treewidth of its primal graph, D the largest variable domain, and $|\mathcal{T}|$ the number of bags in a supplied tree decomposition.

THEOREM 2 (EXACT BOUNDED-WIDTH INFERENCE). *Given a width- τ tree decomposition, both endpoints of any rational factor-additive query can be computed, with extremal witness worlds, in $O(|\mathcal{T}|D^{\tau+1})$ time and $O(|\mathcal{T}|D^\tau)$ space, apart from polynomial arithmetic bit complexity.*

PROOF. Assign each factor to a bag containing its scope. Minimum and maximum messages enumerate at most $D^{\tau+1}$ assignments per bag and retain the best value for each separator assignment. Induction from the leaves proves that each eliminated subtree is optimized conditional on its separator. Root optimization gives the endpoint; stored choices recover a witness world. $\square$

Theorem 2 is a certified tractability parameterization, not a production claim. A KDD-ready system must compile the verified release operator, exploit temporal components, measure the resulting width, and report explicit primal/dual gaps when a component is too wide.

Reference formulation. The implemented coupled reference program uses binary edge variables z_e , assignment variables x_{iv} , and joint edge-label variables y_{euv} only for ordered pairs (u, v) in an edge's declared compatibility set $\mathcal{P}_e$. It enforces Eq. (1), $\sum_{v \in \mathcal{A}_i} x_{iv} = 1$, the count bounds in Eq. (2), and

$$\begin{aligned} \sum_{(u,v) \in \mathcal{P}_e} y_{euv} &= z_e, \\ y_{euv} &\leq x_{iu}, \quad y_{euv} \leq x_{jv}, \\ y_{euv} &\geq z_e + x_{iu} + x_{jv} - 2. \end{aligned}$$

The query is linear in y . Context-only records have assignment variables and enter counts but have no incident selectable edge. An exhaustive enumerator is the exact oracle on small instances; the sparse mixed-integer program is a numerical reference backend. It also enforces optional score floors and supergraph-edge budgets. Exact optima, numerical optima, exact or structural infeasibility, numerical infeasibility, and unresolved runs are reported as distinct states; a timeout is unresolved rather than a separate certificate.

The implementation currently models one categorical label and one count-cell membership per record. The paired pickup/drop-off operator used in Theorem 1 is a theory instance, not a compiled

Table 2: Algorithmic boundary of the declared world model.

Regime	Endpoint computation	Status
Fixed attributes	Weighted core-cover matching	Polynomial baseline
Independent supports	Two edge-envelope matchings	Lemma 1
Paired threshold coupling	Joint labels and matching	NP-complete
Compiled width τ	Min/max factor-graph DP	$O(\mathcal{T} D^{\tau+1})$
General graph	MILP/decomposition	Gap required

Table 3: Machine-readable solver states and their licensed interpretation.

State	Source	Licensed statement
EXACT_OPTIMAL	Exhaustive oracle	Attained endpoint certified
NUMERICALLY_OPTIMAL	HiGHS incumbent	Numerical endpoint; retain gap
PROVEN_INFEASIBLE	Structural check or enumeration	Declared world set empty
NUMERICALLY_INFEASIBLE	HiGHS status	Numerical evidence only
UNRESOLVED	State/time limit or solver failure	No feasibility conclusion

production operator. Multiple endpoint attributes require additional variables and count factors before that theorem can describe an executable application pipeline.

This formulation is not itself the intended KDD algorithmic claim. A scalable certified decomposition must connect the release operator's factor structure to production graphs. Until that gate is passed, the program is a correctness baseline.

5 LEARNING-AUGMENTED RESTRICTIONS

5.1 What Node Outcomes Do Not Identify

Let $\partial M \in \{0, 1\}^V$ be the incidence vector of a matching. A released match indicator can reveal ∂M without revealing M .

PROPOSITION 2 (PARTNER NONIDENTIFICATION). *If a candidate graph contains distinct matchings M_1 and M_2 with $\partial M_1 = \partial M_2$, then node match outcomes and node covariates alone do not identify which of those relations occurred. In particular, an all-positive incidence vector does not distinguish any two perfect matchings.*

PROOF. Take two latent worlds with the same released nodes, covariates, and incidence vector, but relation M_1 in one and M_2 in the other. Their observations coincide while their partner edges differ. No function of the released data can distinguish them without another model or source of relation truth. $\square$

This result does not prohibit structured weak supervision; it fixes its claim boundary. A model may score complete matchings and enforce mutual exclusion, but a node-level objective alone cannot validate partner ranks in the regime of Proposition 2. In

our controlled benchmark, edge scorers are therefore trained on separate synthetic source markets whose relations are observed. Chicago node outcomes are not used as hidden edge labels.

5.2 Calibrating a Matching Set

Let s_e be any scorer fixed before calibration. With optional buffer nodes, the score must use core incidence,

$$S_C(M) = \sum_{e \in M} |e \cap C|_{s_e}. \quad (5)$$

Because $\sum_{e \in M} |e \cap C| = |C|$, replacing s_e by $as_e + b$ for $a > 0$ changes every feasible score by the same additive constant after scaling. Let S_d^- and S_d^+ be the score extrema over the matching projection of market d 's full feasible-world set, including any attribute-conditioned compatibility, and define the true matching's normalized regret

$$R_d = \frac{S_d^+ - S_C(M_d^+)}{S_d^+ - S_d^-}, \quad (6)$$

with $R_d = 0$ for a constant score map. This replaces raw fractions of a score whose zero has no common meaning across models.

Use m calibration markets, disjoint from scorer training, whose true matchings are observed. For target miscoverage α , let $k_\alpha = \lceil (m+1)(1-\alpha) \rceil$ and let $\widehat{\tau}_\alpha$ be the k_α th smallest calibration regret, using the support bound one if $k_\alpha = m+1$. In a new market retain feasible worlds satisfying

$$S_C(M) \geq S^+ - \widehat{\tau}_\alpha(S^+ - S^-). \quad (7)$$

Operationally, the procedure is:

1. train or otherwise fix s_e using source data disjoint from calibration;
2. for each calibration market solve S^- and S^+ over the matching projection of its declared world set and evaluate the observed true matching;
3. take the finite-sample order statistic of the resulting regrets;
4. in a target market impose Eq. (7), then solve the lower and upper program for every predeclared query; and
5. report the radius, candidate-support declaration, witnesses, and solver status together with the query range.

The expensive object is still a structured optimization oracle. Calibration adds one scalar nesting rule; it does not turn edge scores into posterior probabilities.

THEOREM 3 (MARKET-LEVEL CALIBRATED QUERY COVERAGE). *Suppose the scorer and set construction are fixed before calibration, the m calibration markets and one new market are exchangeable, and each true world belongs to its declared candidate and attribute domain. Then*

$$\Pr\{F_{m+1}^\dagger \in \mathcal{F}_{\widehat{\tau}_\alpha}(O_{m+1}, G_{m+1})\} \geq 1 - \alpha.$$

On the same event, the endpoint range over this restricted set simultaneously covers the true value of every predeclared query.

PROOF. Exchangeability makes the new true world's regret rank uniform after random tie-breaking and conservative under deterministic ties. The probability that its rank exceeds k_α is at most α by the definition of k_α . Equation (7) is equivalent to regret at most $\widehat{\tau}_\alpha$. When the true world is retained, applying Proposition 1 to each query yields simultaneous coverage; no additional union

Table 4: Exact declared-world checks in the joint reference implementation.

Intervention	Before	After
Global #A ≤ 2	Independent $[0, 1]$	Coupled $[0, 0.5]$
Label-conditioned edge	$c = A \Rightarrow cx$	$c = B \Rightarrow cy$
Γ relaxation	$\Gamma = 0 : [1, 1]$	$\Gamma = 1 : [0, 1]$
Score restriction	Raw $[0, 1]$	Floor $[1, 1]$
Core-buffer target	Two core incidences	Endpoint 1.0

bound over queries is required because they share the same world-inclusion event. $\square$

The theorem is conditional on candidate support. Given a declared supergraph, we separately allow at most Γ selected edges outside the baseline graph,

$$\sum_{e \in E_{\text{super}} \setminus E_{\text{base}}} z_e \leq \Gamma.$$

If candidate and calibration failures are controlled separately at α_G and α_S , their joint failure is at most $\alpha_G + \alpha_S$. A supplied supergraph or synthetic calibration alone does not establish either rate for Chicago.

The two sensitivities form a nested audit rather than a single opaque knob. Let $\mathcal{F}_{\tau, \Gamma}$ retain worlds with normalized regret at most τ and at most Γ selected supergraph-only edges.

PROPOSITION 3 (NESTED SENSITIVITY PATHS). *If $\tau \leq \tau'$ and $\Gamma \leq \Gamma'$, then $\mathcal{F}_{\tau, \Gamma} \subseteq \mathcal{F}_{\tau', \Gamma'}$. Consequently, for the same query, the lower endpoint cannot increase and the upper endpoint cannot decrease when either budget is relaxed.*

PROOF. Every world satisfying the two tighter inequalities satisfies the two looser ones. Minimization over a superset can only decrease, and maximization can only increase. $\square$

Reporting the (τ, Γ) path exposes whether apparent precision comes from score exclusion or candidate exclusion. It also prevents a calibration claim about τ from being misread as evidence about graph support.

6 CONTROLLED EVALUATION

6.1 Structural and Solver Audit

The first evaluation asks whether the implemented world semantics exhibit the claimed interactions before any learned score is used. Table 4 records exact, deterministic cases. The four-record count example is the worked construction above; the label-conditioned case changes the selected partner when a global count fixes the core label. A separate seeded audit generates 250 complete four- or six-core instances with random supports, count bounds, compatibility tables, and Γ budgets. Exhaustive and HiGHS feasibility statuses agree on all 250; endpoints agree on all 185 feasible cases, while 65 are infeasible. The exhaustive side supplies the certificate; agreement does not make the floating-point side exact.

The artifact currently has 46 passing unit tests and eight executable adversarial checks. These cover input validation, state-limit semantics, positive-affine score invariance, nonattainable interval

interiors, signed objectives, and node-marginal nonidentification. They support software behavior for declared finite inputs, not the correctness of an external observation operator.

6.2 Matching-Set Calibration under Deliberate Query Leakage

Design. We use a deterministic, disjoint source/calibration/test split. Forty source markets directly supervise logistic edge scorers with observed true edges; no node-label learner is evaluated here. Forty-nine target-style markets calibrate Eq. (6), and 120 held-out markets are used for final evaluation. Each market has six hidden pairs, fully observed binary endpoint attributes, and a complete candidate graph. There is no privacy-count coupling in this experiment, so it isolates the matching-set calibration layer.

Target-free features measure route, time, and duration similarity. A query-leaking diagnostic additionally sees the same-SES edge contribution that defines the downstream query. The source homophily generating probability is 95%, while calibration and test use 55%, deliberately shifting that circular feature. The nominal matching-set coverage is 90%. Radius 0.05 is an illustrative stress point, not a selected or calibrated baseline. The committed program uses numerical HiGHS endpoints; independent exhaustive enumeration over all 10,395 perfect matchings per market reproduces the reported counts and widths.

Result. The query-leaking radius-0.05 stress point looks strongest if evaluated only by width: it removes 95.8% of the score-free ambiguity. It retains 18/120 true matchings and covers the downstream query in 29/120 markets. Calibration detects the shifted shortcut. Its radius expands to 0.504, retaining 117/120 true matchings and covering the scalar query in 120/120 markets, while width reduction falls to 7.6%. The target-free score needs a smaller radius but also gains little calibrated precision. This single seed and market size do not establish robustness. The result is narrower: calibration makes the coverage price of sharpening measurable rather than rewarding circular precision.

Software audit. The benchmark validates matching-level calibration only. It does not validate node weak supervision, latent-attribute inference, privacy-count coupling, candidate support, or transfer to Chicago. Multi-seed, size, sparsity, coarsening, and candidate-omission sweeps remain required evidence.

7 CHICAGO FLAGSHIP APPLICATION: EVIDENCE GATE

Chicago’s public Transportation Network Provider table reports whether each customer transaction shared a vehicle with a separately booked customer and reports the pooled chain size [2]. It does not publish the partner or shared-trip identifier. Conditional on boundary completeness and reporting consistency, matched $K = 2$ transactions therefore induce a hidden nonbipartite perfect matching. The application target is the neighborhood-context composition of these realized two-transaction matches.

The production analysis is not yet licensed. The committed data are a deterministic two-character trip-ID-prefix sample. Conditional on retaining one member of a two-record relation, that design omits

the counterpart with probability approximately 255/256 under independent prefix retention. It is suitable for schema checks, not partner reconstruction or endpoint bounds. Moreover, the accessible historical tract-suppression clarification is tied to legacy datasets; its unchanged operation in the 2025/2026 release has not been verified. We therefore do not encode the historical rule as current fact.

The complete application requires all transactions contributing to each privacy cell, boundary-buffered matched rows, a necessary-condition candidate supergraph, the correct tract vintage, and full solver status by component. Primary comparisons will include complete cases, community-area aggregation, independent attribute envelopes, one point imputation, spatiotemporal proxy grouping, score-free coupled endpoints, and calibrated restrictions only where independent partner truth supports them. No Chicago composition or January 2026 policy sign is reported in this draft.

8 LIMITATIONS AND ETHICAL CONSIDERATIONS

The endpoint guarantee is conditional on the declared world set. Candidate omissions, an incorrect release operator, nonexchangeable calibration markets, or unmodeled attribute supports can remove truth before optimization. The split-conformal result is marginal across complete markets, not conditional on every market and not transferable from simulation by assertion. Numerical MILP optima are not exact mathematical certificates; production reporting must retain primal/dual gaps and unresolved states.

The Chicago unit is a customer transaction, not a stable person. Census attributes describe neighborhood context, not rider income or race. Releasing one extremal latent assignment could facilitate reidentification, so the intended artifact reports only aggregate endpoints, support diagnostics, and solver certificates. It must not publish imputed partner links, trip IDs, or latent fine-geography assignments. Institutional review and the City’s data terms must be resolved before any nonpublic data are used.

9 GENERATIVE AI USAGE

Generative AI assisted code drafting, adversarial counterexample search, and language editing. Every mathematical claim, experiment, citation, and code path remains subject to author verification. No generated text or synthetic output is treated as evidence about Chicago.

10 CONCLUSION

Privacy coarsening can hide a relation and its endpoint attributes at the same time. The appropriate output is then a query over feasible worlds, not a preferred reconstructed graph. Independent attribute supports reduce to weighted matching; global release coupling creates the substantive new problem. Matching-level conformal calibration can prevent a learned scorer from receiving credit for false precision, but it may also reveal that little safe sharpening is available. The remaining work is decisive rather than cosmetic: verify the current operator, turn the bounded-width route into a production certified algorithm, validate on another ground-truth relation, and run the full Chicago application.

Table 5: Fixed held-out synthetic markets: coverage cannot be read from width alone.

Scorer	Calibrated restriction			Arbitrary radius 0.05			
	Match cov.	Query cov.	Width red.	Match cov.	Query cov.	Width red.	Point MAE
Target-free	95.8%	100%	3.8%	30.0%	83.3%	70.6%	0.122
Query-leaking diagnostic	97.5%	100%	7.6%	15.0%	24.2%	95.8%	0.308

The score-free range covers the query in every test market and has mean width 0.661. Calibrated radii are 0.341 (target-free) and 0.504 (query-leaking). “Match cov.” tests whether the entire hidden matching remains eligible; “Query cov.” is weaker because an excluded matching’s scalar value can still fall between the endpoints. The observed 100% values mean 120/120, not a strengthened theoretical coverage level.

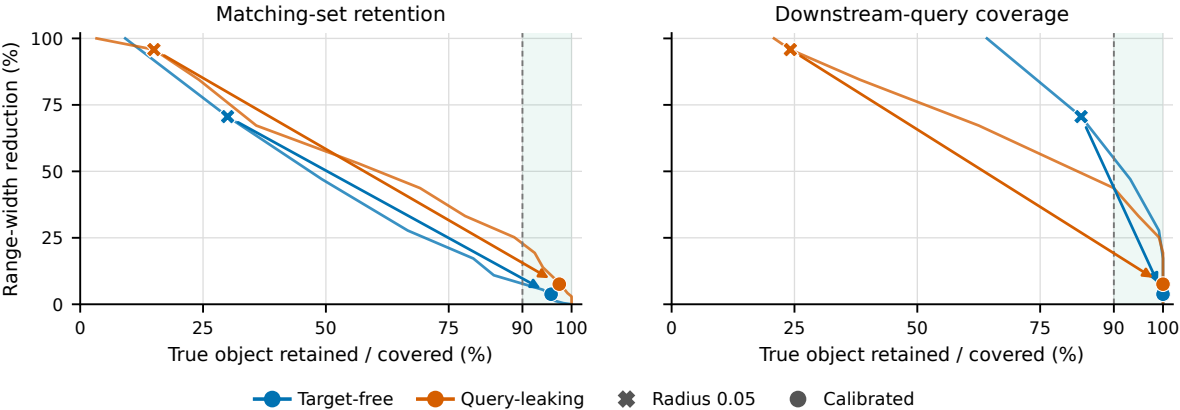


Figure 2: The illustrative tight radius occupies the upper-left region: large width reduction but poor retention or coverage. Market-level calibration moves both scorers across the nominal 90% line, but most apparent precision disappears. Thin paths show the exact-enumeration radius grid from 0 to 1; arrows connect the radius-0.05 stress point to the calibrated radius for the same scorer.

Table 6: Chicago observation-operator contract in this working draft.

Status	Object and licensed use
Released	Transaction-level shared-match flag and reported chain size
Latent	Which two eligible transactions are partners; suppressed fine context
Unverified	Current threshold logic, contributing-cell population, boundary support
Prefix sample	Schema and mechanics checks only
Full extraction	Mechanism-conditional contextual-composition endpoints, if all gates pass
Never licensed	Person-level SES, preferences, durable ties, or inferred row-level links

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A REPRODUCIBILITY AND CLAIM STATUS

The artifact includes the complete benchmark generator, split-conformal calibrator, exact small-instance matching oracle, sparse numerical reference solver, fixed seeds, result tables, and executable adversarial examples. Hidden pair truth is read only in the source-training, calibration, or held-out-test stage declared by the benchmark split. The earlier two-day weak-node-score pilot is retained as failure-analysis provenance but is not a headline result.

Current completed claims are limited to Propositions 1, 2, and 3, Lemmas 1 and 2, Theorems 1, 2, and 3, 46 passing declared-input unit tests, eight adversarial checks, the seeded 250-instance solver audit, and the controlled results in Table 5. The scalable decomposition, current Chicago observation operator, non-Chicago relation benchmark, and complete-day Chicago estimates remain open gates and cannot be inferred from the appendix.

B FULL PROOF OF THEOREM 1

Define the decision problem as follows. Records have pickup and drop-off time labels, finite attribute supports, and an observed symbol that is either a visible attribute pair or missing. A world chooses both attributes. The paired operator releases both exactly when both selected endpoint cells have count at least k . Target records form left and right sets and must be perfectly matched; context records cannot be matched. An explicitly represented table $\Gamma_{xy}(p_x)$ determines compatibility. A certificate is polynomially checked by

computing all counts and matching degrees, so the problem is in NP.

Reduce perfect 3D matching with $|X| = |Y| = |Z| = q$. If needed, add one disjoint forced triple so $q \geq 2$ without changing the answer. Create target records r_x and r_y and the complete outer graph $X \times Y$. All r_x share one pickup time and have support Z . Each r_x has a private drop-off time and a singleton drop-off support; its observed pair is missing. Each r_y has private endpoint times, singleton supports, and a missing observed pair.

For each $z \in Z$, create $k - 1$ context anchors with the common pickup time and singleton pickup support $\{z\}$. All anchors share a drop-off time and a singleton label unused by targets, and are visibly released. The anchor drop-off count is $q(k - 1) \geq k$. If $n_z = |\{x : p_x = z\}|$, each z -anchor pickup count is $(k - 1) + n_z$; visibility is therefore equivalent to $n_z \geq 1$. Because $\sum_z n_z = q$, every $n_z = 1$. Missing target outputs are consistent: every x has a unique count-one drop-off cell, and every y has unique count-one endpoint cells.

Finally set $\Gamma_{xy}(z) = 1$ exactly for input triples (x, y, z) . A feasible world matches every x and y once and assigns every z once, so its selected triples form a perfect 3D matching. Conversely, a perfect 3D matching selects the corresponding p_x and (x, y) edges; all anchor counts equal or exceed k , all targets remain suppressed, and every selected edge is compatible. For fixed k the construction has $O(q)$ records, $O(q^2)$ support and outer-edge entries, and at most an $O(q^3)$ explicit compatibility table. The reduction is polynomial.